

Year-Dependent YKS Score Volatility and Unfair Outcomes Under the Bavarian Grade Conversion Formula

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Abstract. This project investigates a structural issue arising when absolute exam scores from different years are converted into German grades using the modified Bavarian grade conversion formula. Using publicly available YKS admission data for Computer Engineering programs from 2021 and 2022, the project combines automated data scraping, data cleaning, and visualization to analyze the relationship between admission ranks and placement scores across years. The results show that while admission ranks remain relatively stable for comparable programs, absolute scores exhibit substantial year-dependent shifts. As a consequence, absolute-score-based grade conversion may lead to systematically different outcomes for applicants with similar relative performance. The project was carried out as part of the course and emphasizes reproducible data collection and analysis using Python-based automation.

Keywords: YKS, grade conversion, NC admission, data scraping, visualization, automation

1 Introduction

Admission to NC-restricted degree programs in Germany (*NC-beschränkt*) often relies on converting foreign educational achievements into the German grading scale. For many international applicants, including those presenting Turkish YKS outcomes, institutions and services frequently rely on the (*modified*) *Bavarian grade conversion formula*, which maps a foreign result on an absolute scale into the German range $[1.0, 4.0]$, where 1.0 is best.

This paper studies a structural issue that arises when an absolute-score-based conversion is applied to YKS results across different exam years. The Turkish YKS produces both (i) an absolute placement score (used for admissions in Turkey), and (ii) a nationwide rank (Başarı Sıralaması), which is a relative measure of performance within the cohort. If exam difficulty changes across years, then absolute scores can shift substantially even when ranks remain comparatively stable for the same programs and institutions. Consequently, applicants of similar relative strength (similar rank) can receive different German converted grades solely due to the exam year, which can affect admission decisions in NC-restricted contexts.

Contribution. Using scraped public admission tables for Computer Engineering (*Bilgisayar Mühendisliği*) for YKS 2021 and YKS 2022, we quantify the mismatch between rank stability and score volatility, and we show how the Bavarian formula, when fed with absolute YKS scores, can induce year-dependent unfairness.

2 Background: YKS Placement Score and Its Components

The YKS placement score used for university placement in Turkey is a composite quantity derived from exam components and a high-school achievement contribution.

2.1 Exam components: TYT and AYT

Let T denote the TYT-derived contribution and A the AYT-derived contribution. For many quantitative tracks (e.g., SAY), the placement score uses weights:

$$S_{\text{exam}} = 0.4T + 0.6A. \quad (1)$$

2.2 High-school achievement: OBP

Let $\text{OBP} \in [0, 100]$ denote the high-school achievement score. The additional placement contribution is

$$S_{\text{OBP}} = 0.6 \cdot \text{OBP}, \quad (2)$$

so that $S_{\text{OBP}} \in [0, 60]$ and the total placement score is

$$S = S_{\text{exam}} + S_{\text{OBP}}. \quad (3)$$

2.3 Absolute score vs. relative rank

For each exam year y , the system yields:

- an absolute score $S^{(y)} \in \mathbb{R}$ (placement score), and
- a nationwide rank $R^{(y)} \in \mathbb{N}$ (lower is better).

Intuitively, $R^{(y)}$ is a cohort-relative statistic, while $S^{(y)}$ is cohort-dependent and can shift with exam difficulty, scaling, and normalization procedures.

3 The Bavarian Grade Conversion Formula

German institutions frequently refer to the *modified Bavarian formula* to convert a grade/score from a foreign system to the German grading scale. A commonly stated form is:¹

$$G = 1 + 3 \cdot \frac{P_{\max} - P}{P_{\max} - P_{\min}}, \quad (4)$$

where

- G is the German grade (best = 1.0, minimum passing = 4.0),
- P is the applicant's result in the original system,
- $P_{\max}$ is the best achievable value in the original system,
- $P_{\min}$ is the minimum passing value in the original system.

The formula appears in institutional explanations and in KMK-related documentation.[1,2]

3.1 Key sensitivity property

A crucial feature of Eq. (4) is *linearity in the absolute score P* . Therefore, any year-dependent shift in P (even when rank is stable) induces a year-dependent shift in G .

Proposition 1 (Linear year-sensitivity). *Fix $P_{\max}$ and $P_{\min}$. For two applicants i and j with results P_i and P_j , the difference in converted German grades is*

$$G_i - G_j = 3 \cdot \frac{P_j - P_i}{P_{\max} - P_{\min}}. \quad (5)$$

Hence, if an exam year produces systematically lower absolute scores for the same rank region, then the converted grades will systematically worsen.

3.2 Fairness criterion

We formalize the intuitive fairness goal:

Definition 1 (Rank-consistency across years). *A conversion pipeline is rank-consistent across years if, whenever two applicants have comparable rank (relative performance) across years, their converted grades are comparable up to a small tolerance that does not depend on the year.*

When only $P = S^{(y)}$ (absolute YKS score) is used, rank-consistency can fail if $S^{(y)}$ drifts with year-specific difficulty.

¹ Many institutions provide calculators and descriptions of the formula; see, e.g., TUM and KMK references.

4 Data Sources and Scraping Methodology

4.1 Public admission tables

We collect program-level admission outcomes for Computer Engineering (SAY track) for 2021 and 2022 from publicly accessible tables:

- basarisiralamalari.com (tables containing rank and score entries)[3]
- kazanabilirsin.com (tables containing year-sliced rank/score entries)[4]

Each row contains (when available): university name, program variant (e.g., scholarship tiers), score type, quota/placed counts, last-admitted rank, and last-admitted score.

4.2 Scraping and cleaning

Two Python scripts were used to:

- download HTML pages with `requests`,
- parse table structures with `BeautifulSoup`,
- normalize numeric formats (e.g., thousand separators, decimal commas),
- export structured CSVs using `pandas`.

Duplicates were removed by the key (university, program, score_type) to prevent repeated entries from multi-table layouts.

4.3 Matched-year comparison setup

Let $\mathcal{U}$ denote the set of university-program entries that can be matched across years (string-normalized). For each matched entry $u \in \mathcal{U}$ and year $y \in \{2021, 2022\}$ we denote:

$$(R_u^{(y)}, S_u^{(y)}) \text{ as the last-admitted rank and last-admitted score.} \quad (6)$$

We analyze:

- **Rank stability:** how close $R_u^{(2021)}$ and $R_u^{(2022)}$ are,
- **Score volatility:** how large $|S_u^{(2021)} - S_u^{(2022)}|$ is,
- **Conversion gap:** the implied difference in G if Eq. (4) is applied on scores.

5 Empirical Analysis: Rank Stability vs. Score Volatility

This section reports the main empirical pattern: for many comparable programs, admission ranks are relatively stable across 2021 and 2022, but admission scores shift noticeably.

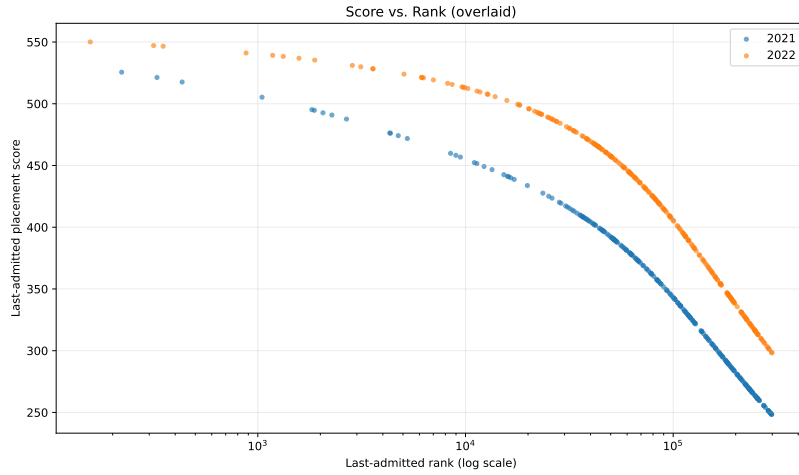


Fig. 1: Scatter plot of last-admitted placement score S versus last-admitted rank R for YKS 2021 and YKS 2022 (overlaid, log-scaled rank axis). The plot illustrates how absolute scores can shift between years even in similar rank regions.

5.1 Overview and matched comparisons

We begin by contrasting absolute placement scores with relative admission ranks. Figure 1 provides an overview of the score–rank relationship for both years, while Figures 2a and 2b isolate rank stability and score volatility, respectively, using matched program entries.

5.2 Distributional evidence: systematic vs. isolated effects

To assess whether the observed differences are isolated outliers or a systematic phenomenon, we examine the distributions of rank changes and score changes across all matched programs.

5.3 Institution-level illustration and grade-implication

While the previous figures summarize aggregate patterns, we now focus on selected universities to illustrate the effect at an institutional level.

Finally, we translate score differences into implied German grade differences using the Bavarian conversion formula, purely for illustrative purposes.

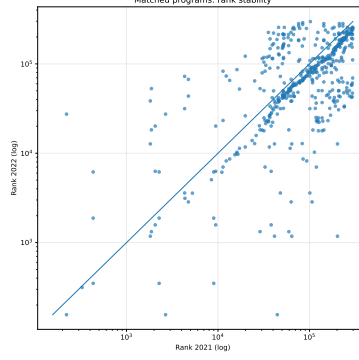
5.4 Interpretation of the core pattern

If Fig. 2a concentrates near the identity line while Fig. 2b shows wider dispersion, this supports the empirical claim:

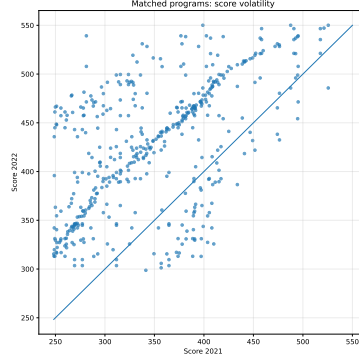
$$R_u^{(2021)} \approx R_u^{(2022)} \quad \text{but} \quad S_u^{(2021)} \not\approx S_u^{(2022)}. \quad (7)$$

Under an absolute-score conversion (Eq. (4)), this implies year-dependent differences in German grades even for similar relative standing.

Fig. 2: Matched-entry comparisons for admission ranks and admission scores.



(a) Matched-entry comparison of admission ranks. Each point corresponds to a matched program u , plotted as $(R_u^{(2021)}, R_u^{(2022)})$ with the identity line. Concentration near the diagonal indicates rank stability across years.



(b) Matched-entry comparison of admission scores. Each point corresponds to a matched program u , plotted as $(S_u^{(2021)}, S_u^{(2022)})$ with the identity line. Dispersion away from the diagonal indicates substantial year-dependent score shifts.

6 Implications for NC-Restricted Admission in Germany

Assume an applicant is assessed using the converted grade G derived from $P = S^{(y)}$. Consider two applicants a and b from different years with comparable ranks:

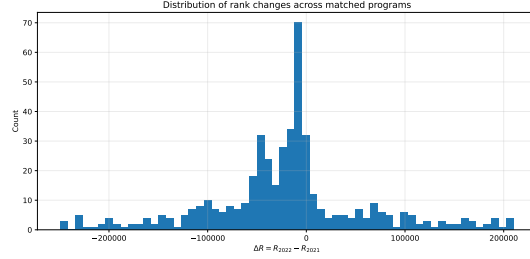
$$R_a^{(2021)} \approx R_b^{(2022)}. \quad (8)$$

If the corresponding absolute scores differ by $\delta = S_b^{(2022)} - S_a^{(2021)}$, then Eq. (4) yields a converted-grade difference

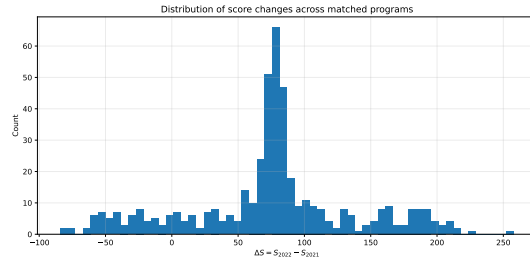
$$\Delta G = G_b - G_a = -3 \cdot \frac{\delta}{P_{\max} - P_{\min}}. \quad (9)$$

Thus, even when ranks indicate comparable performance, absolute-score differences can translate into meaningful grade gaps. In NC-restricted programs where selection thresholds can be tight, such a gap may change an admission outcome.

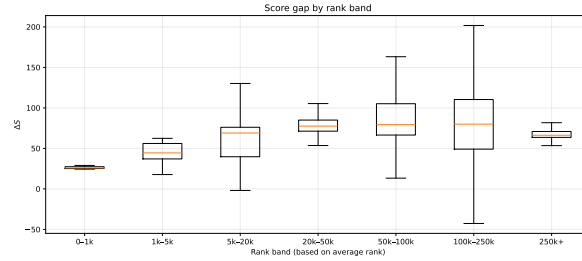
Fig. 3: Distributions of matched-program rank and score changes, and score changes by rank band.



(a) Distribution of rank differences $\Delta R_u = R_u^{(2022)} - R_u^{(2021)}$ across matched programs. A concentration around zero indicates limited rank movement between years.



(b) Distribution of score differences $\Delta S_u = S_u^{(2022)} - S_u^{(2021)}$ across matched programs. Compared to Fig. 3a, the wider spread highlights substantial score volatility.



(c) Score differences ΔS_u grouped by average rank bands. The boxplots show that score volatility persists across multiple rank intervals, including top-ranked programs.

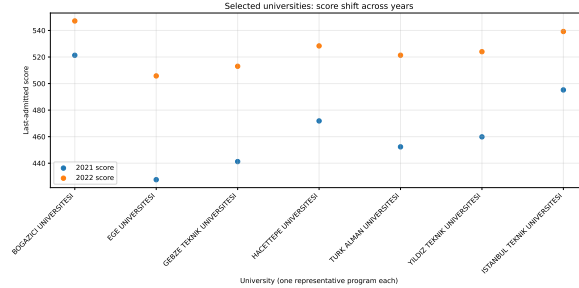
7 Privacy Note for the Public Copy

The original course submission included individual ÖSYM result documents as illustrative examples. These documents have been omitted from this public version to protect personal data. The program-level analysis and all reproducible figures remain unchanged.

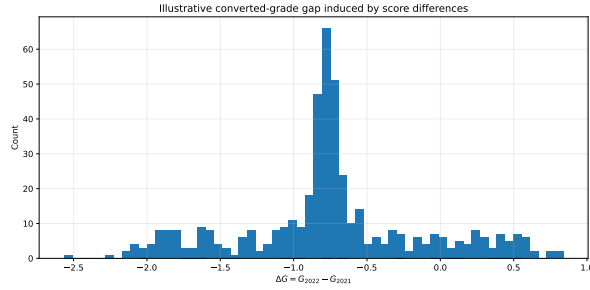
8 Discussion and Recommendations

The empirical claim investigated here is not that YKS is unfair internally, but that *cross-year comparability* of absolute scores is limited when exam difficulty and scaling vary. When a German conversion pipeline ignores ranks and relies

Fig. 4: Selected university illustration and implied grade differences under Bavarian conversion.



(a) Selected university case study. For each university, one representative Computer Engineering program is shown, comparing last-admitted scores for 2021 and 2022 side-by-side.



(b) Illustrative distribution of implied German grade differences ΔG_u resulting from applying the Bavarian formula to year-dependent YKS scores. The figure demonstrates how score shifts alone can induce non-negligible grade gaps.

solely on absolute scores, it implicitly assumes cross-year score comparability. The matched-program evidence challenges that assumption.

A principled mitigation is to incorporate cohort-relative information. Two examples:

- **Rank-based normalization:** convert via percentile or rank bands instead of raw points.
- **Year-aware calibration:** use a function $f_y(\cdot)$ so that $f_y(S^{(y)})$ becomes comparable across years before applying Eq. (4).

Such approaches target rank-consistency across years by design.

9 Relation to Course Topics

Table 1 summarizes how the topics covered in the course were applied within this project. The focus was placed on automation, data processing, and visualization, which were used in depth, while other topics were employed only as supporting tools or not at all.

Table 1: Relation between course topics and project implementation

Course Topic	Usage in the Project
Linux	The project was developed and executed in a Linux environment, which served as the execution and development platform but was not itself the subject of analysis.
Text Editors	Used for writing and maintaining source code and documentation; not a focus of the project.
Git	Used for version control throughout the development of the project repository.
Docker	Not used in this project.
Automation (Python)	Used in depth. Python scripts were written to automatically scrape admission data from public websites, clean and normalize the data, and export it into structured formats.
Web / HTTP	Used in depth through automated retrieval and parsing of HTML content from public admission tables.
Data Processing	Used in depth. Collected data were cleaned, normalized, matched across years, and analyzed to study rank stability and score volatility.
Visualization	Used in depth. Multiple figures were generated programmatically to visualize relationships between ranks, scores, and derived grade differences.
L ^A T _E X	Used to produce the final project report in a structured and reproducible academic format.
LLMs	Used only marginally as a support tool, not as a core component of the project.

10 Conclusion

This paper frames and empirically investigates a year-dependent unfairness risk when Turkish YKS outcomes are converted into German grades using an absolute-score-based Bavarian formula. The key mechanism is the divergence between (i) relatively stable admission ranks for comparable programs and (ii) volatile absolute admission scores across exam years. Because the Bavarian formula is linear in the absolute score, it can translate year-driven score shifts into materially different converted grades, potentially impacting decisions in NC-restricted admissions.

Future work includes expanding beyond a single field (Computer Engineering) and formally estimating year-calibration mappings using larger matched datasets.

References

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